

JKBOSE 2026

CLASS 11TH

Plant Growth and Development

LECTURE-02

BOTANY

ARSHID BHAT SIR



Topics To Covered

- 1** Quantitative Comparison Of Growth
- 2** Conditions For Growth
- 3** Development
- 4** Differentiation, De-differentiation , Re-differentiation
- 5** Plant Growth Regulators

Quantitative Comparison Of Growth

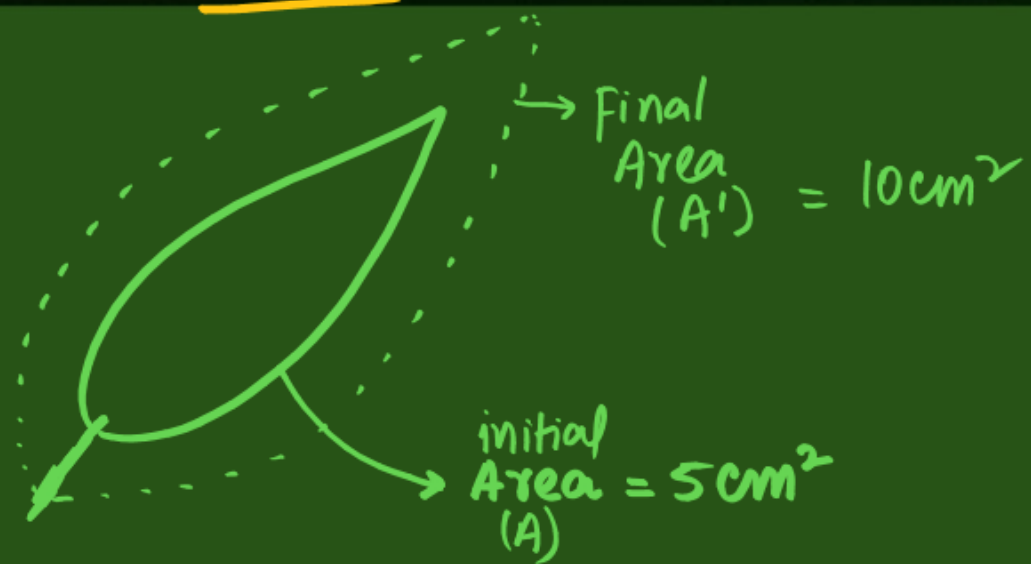
Quantitative comparison of growth of different living organisms can be done by 02 ways:

① Absolute growth rate = $\frac{\text{Total growth}}{\text{Unit time}}$

* Total growth = Final - Initial.

② Relative growth rate = $\frac{\text{Total growth per unit time}}{\text{Initial size}} \times 100$

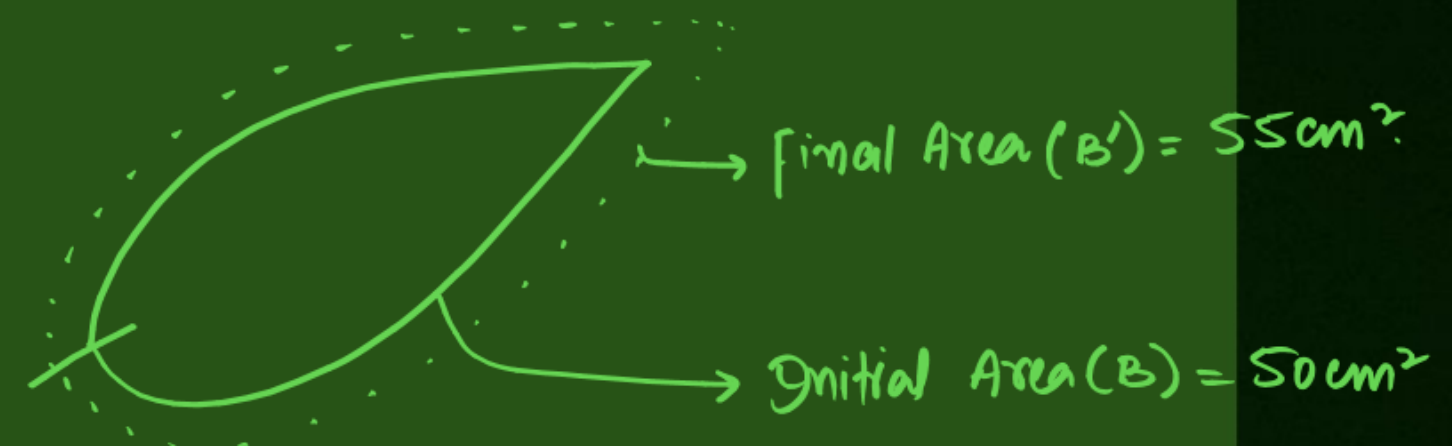
A LEAF



$$\begin{aligned} \rightarrow \text{Absolute Growth rate} &= \frac{\text{Total Growth}}{\text{unit time}} \\ &= \frac{10 \text{ cm}^2 - 5 \text{ cm}^2}{1} = 5 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Relative Growth rate} &= \frac{\text{Total Growth per unit time}}{\text{Initial Size}} \times 100 \\ &= \frac{5 \text{ cm}^2}{5 \text{ cm}^2} \times 100 = 100\% \end{aligned}$$

B - LEAF



$$\begin{aligned} \rightarrow \text{Absolute Growth rate} &= \frac{55 \text{ cm}^2 - 50 \text{ cm}^2}{1} = 5 \text{ cm}^2 \end{aligned}$$

$$\rightarrow \text{Relative Growth rate} = \frac{5 \text{ cm}^2}{50 \text{ cm}^2} \times \frac{100}{1} = 10\%$$

Conclusion: 1. Absolute Growth rate of A & B are same.
2. Relative Growth rate of A > B

Conditions For Growth

Conditions for Growth $\begin{cases} \rightarrow \text{External Conditions.} \\ \rightarrow \text{Internal Conditions.} \end{cases}$

EXTERNAL CONDITIONS

- ① Water: It is one of the reactants of photosynthesis.
 - Acts as a medium for various metabolic reactions.
 - It causes cell enlargement.
- ② Oxygen: It is required for aerobic respiration which forms ATP - required for growth.

- ③ Nutrients: Absorbed from Soil.
 - $\rightarrow$ Required for growth.
 - $\rightarrow$ Both micro & macro-nutrients.

- ④ Temp.: Optimum temperature required for growth.

- ⑤ Sunlight: $\rightarrow$ Required in photosynthesis.
 - $\rightarrow$ PAR (400-700 nm).

INTERNAL CONDITIONS

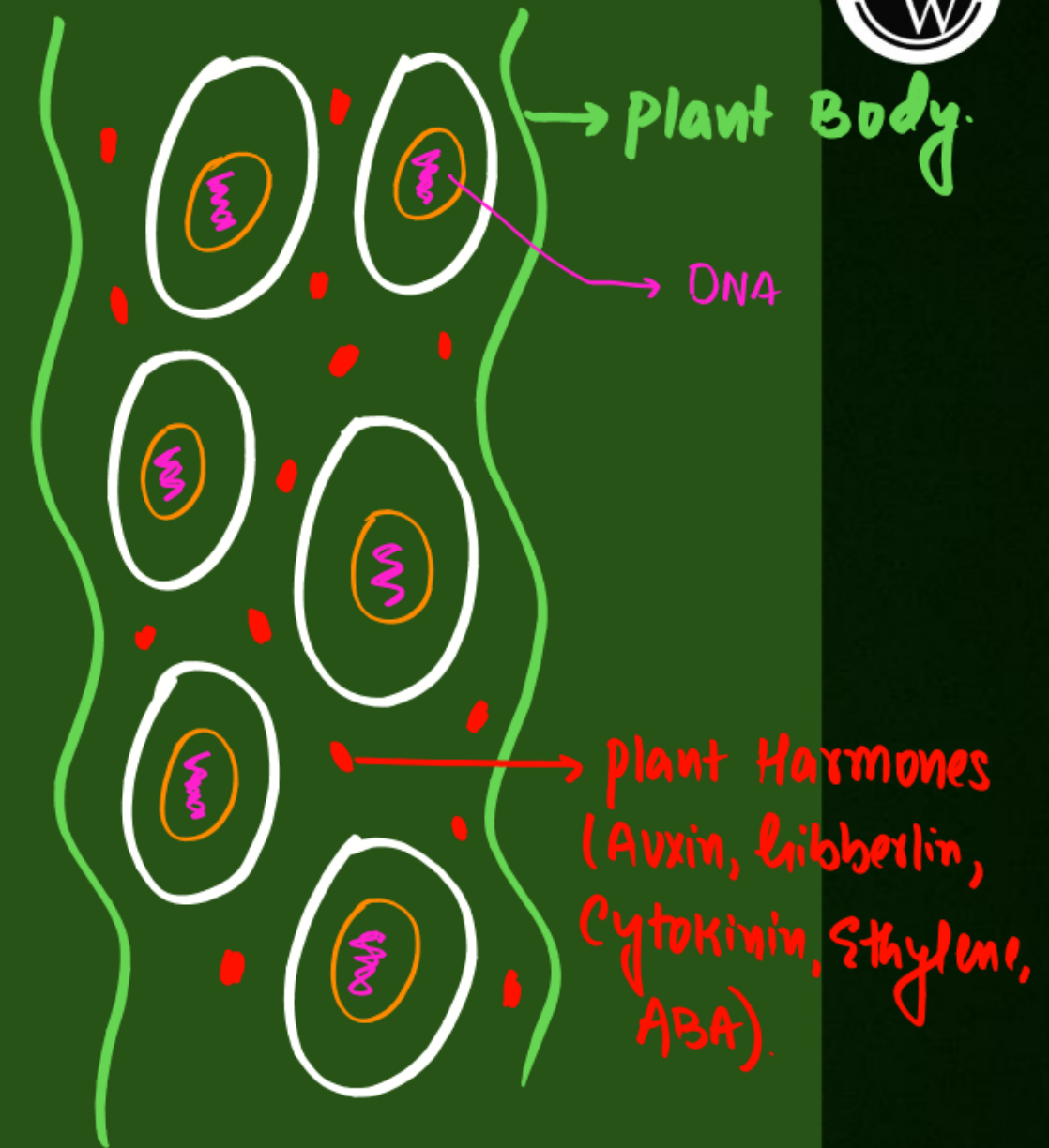
a. Intrinsic intracellular
Factor

Example: DNA

Intrinsic intercellular
Factors

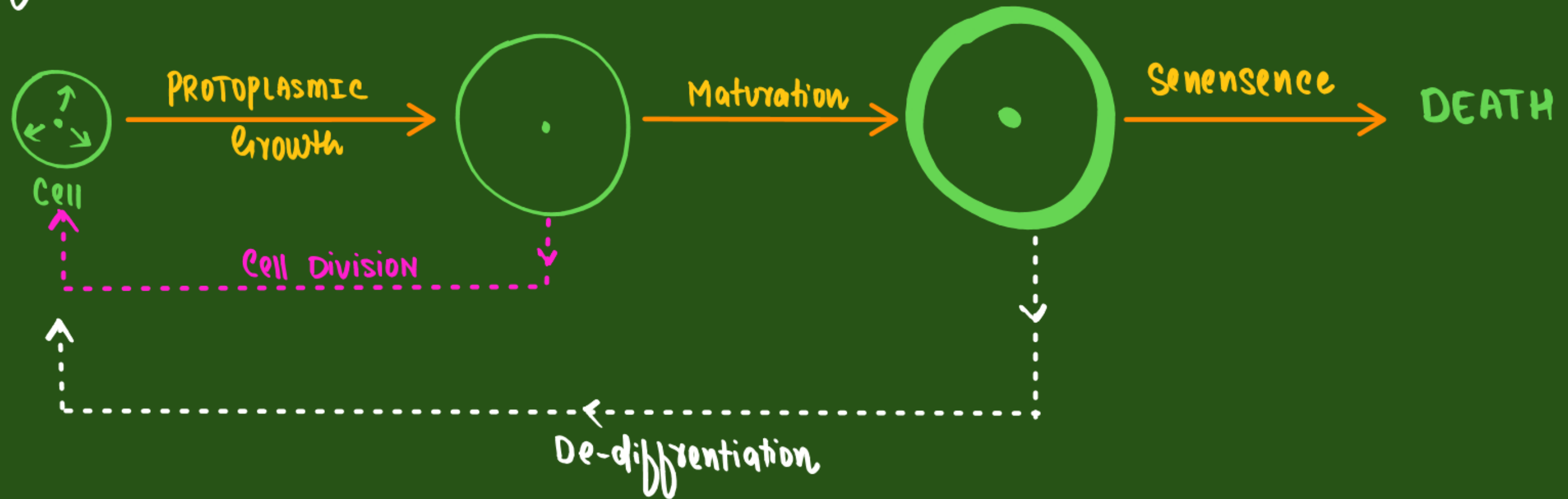
Ex. Plant Growth Regulators

↓
Auxin
Gibberellin
Cytokinin
Ethylene
ABA.



DEVELOPMENT

- It is Sum total of growth and differentiation.
- Development includes all the changes an organism goes through during its life cycle from seed germination upto Death.



NOTE Plant follows different pathways in response to different Environmental Conditions. This ability of Plants is kld as **Plasticity**

• Example of plasticity is **Heterophilly** → presence of different kinds of leaves in same plant species.

• Heterophilly — 02 types:

a. **Environmental heterophilly**

• difference in leaf shape because of different environmental conditions like one in water and 2nd in terrestrial. Ex. Buttercup.

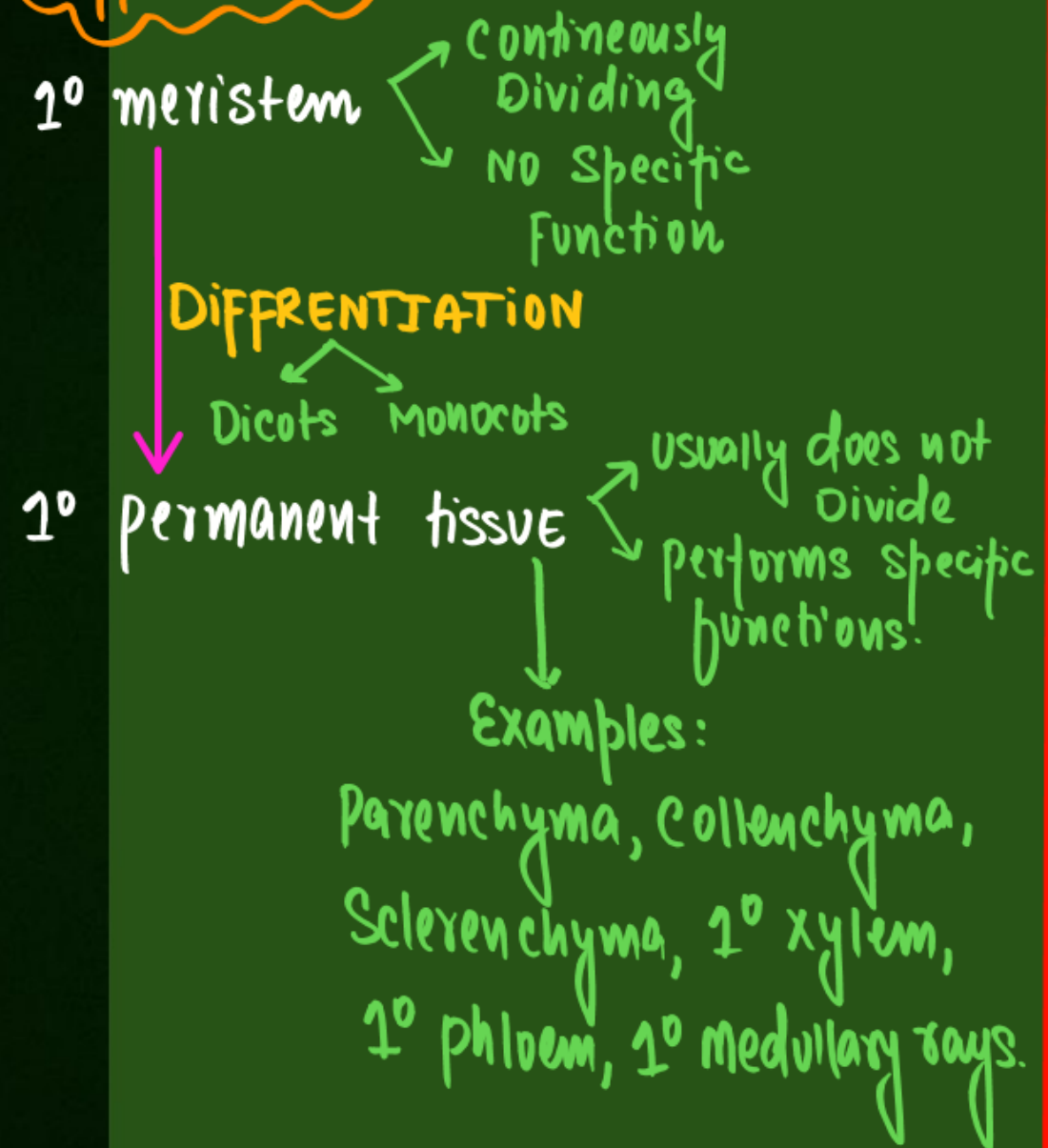
b. **Developmental heterophilly**

• leaves of juvenile plant are different from leaves of Adult plant.
Ex. Cotton, Coriander, Larkspur.

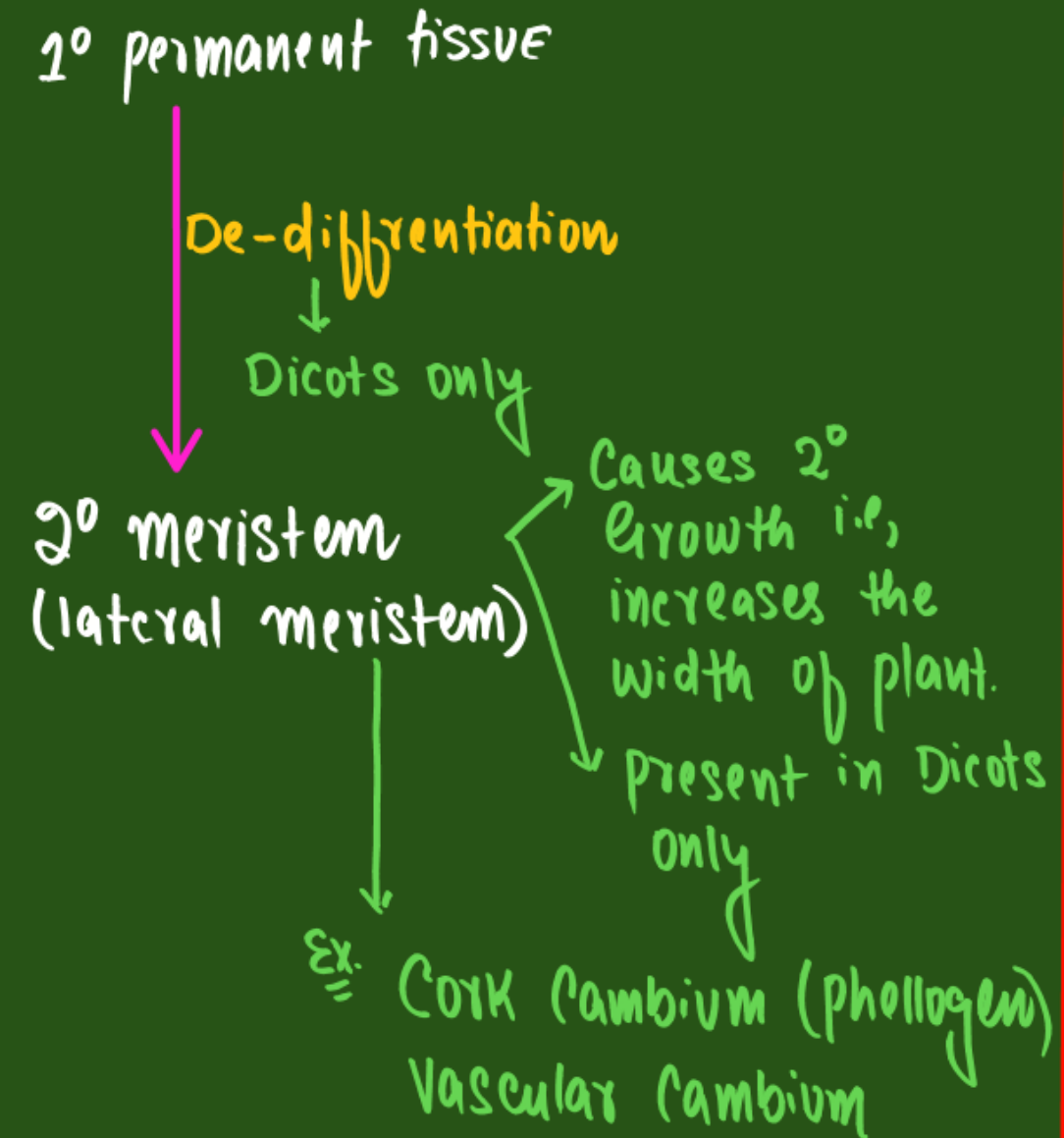
Differentiation, De-differentiation, Re-differentiation



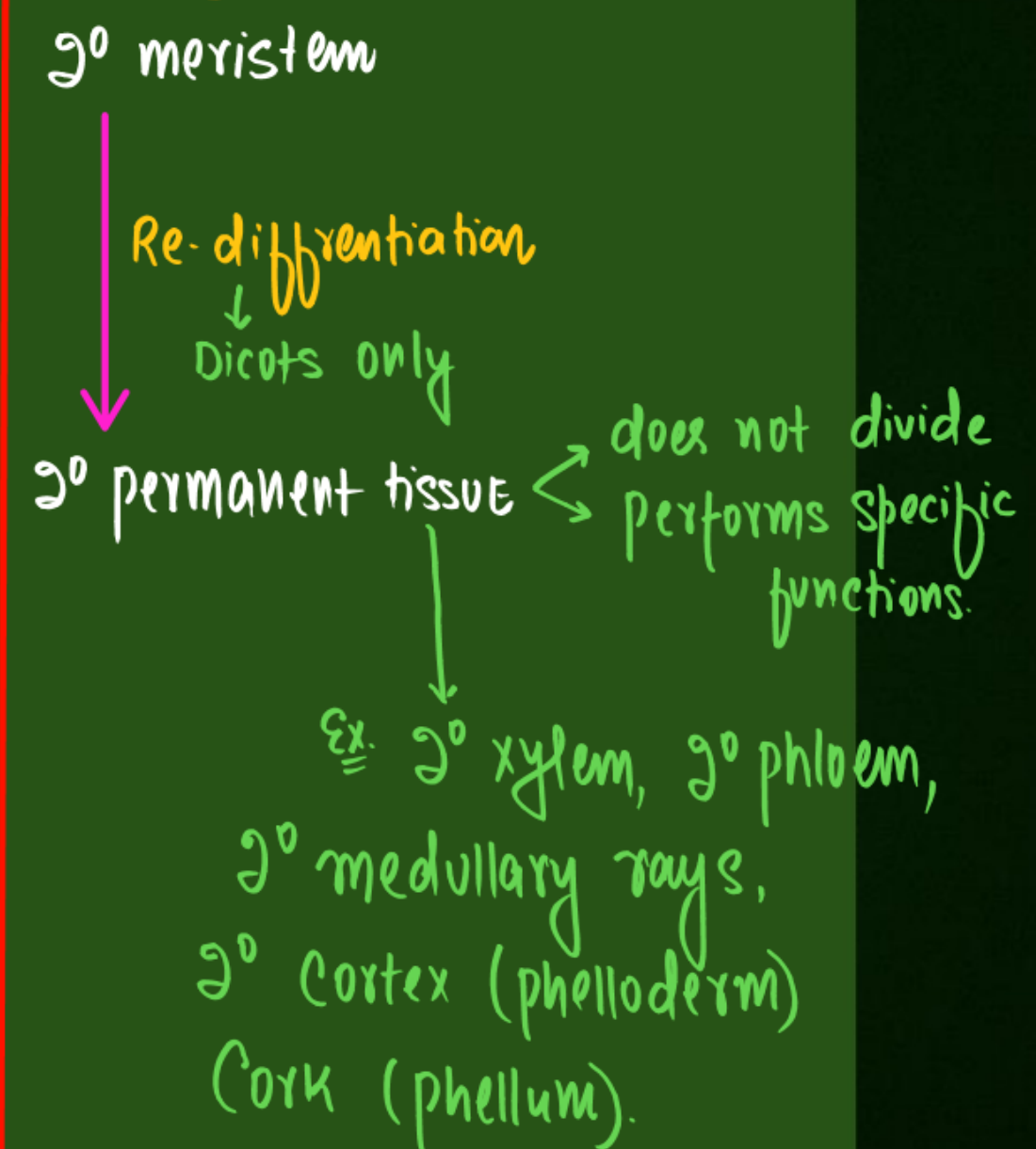
Differentiation



De-differentiation



Re-differentiation



Plant Growth Regulators

↓
Next lecture....

QUESTIONS

The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called: **(2022 Re)**

- 1 Differentiation
- 2 Redifferentiation
- 3 Development
- ✓ 4 Plasticity

QUESTIONS

Which one of the following plants does not show plasticity?

(2022)

- ☒ 1 Maize
- ☐ 2 Cotton
- ☐ 3 Coriander
- ☐ 4 Buttercup

QUESTIONS

Formation of interfascicular cambium from fully developed parenchyma cells is an example for (2024)

- ✓ 1 Dedifferentiation
- 2 Maturation
- 3 Differentiation
- 4 Redifferentiation



Thank
You